

B30801 Managerial Accounting Autumn 2025

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Session #1B: Cost Concepts.

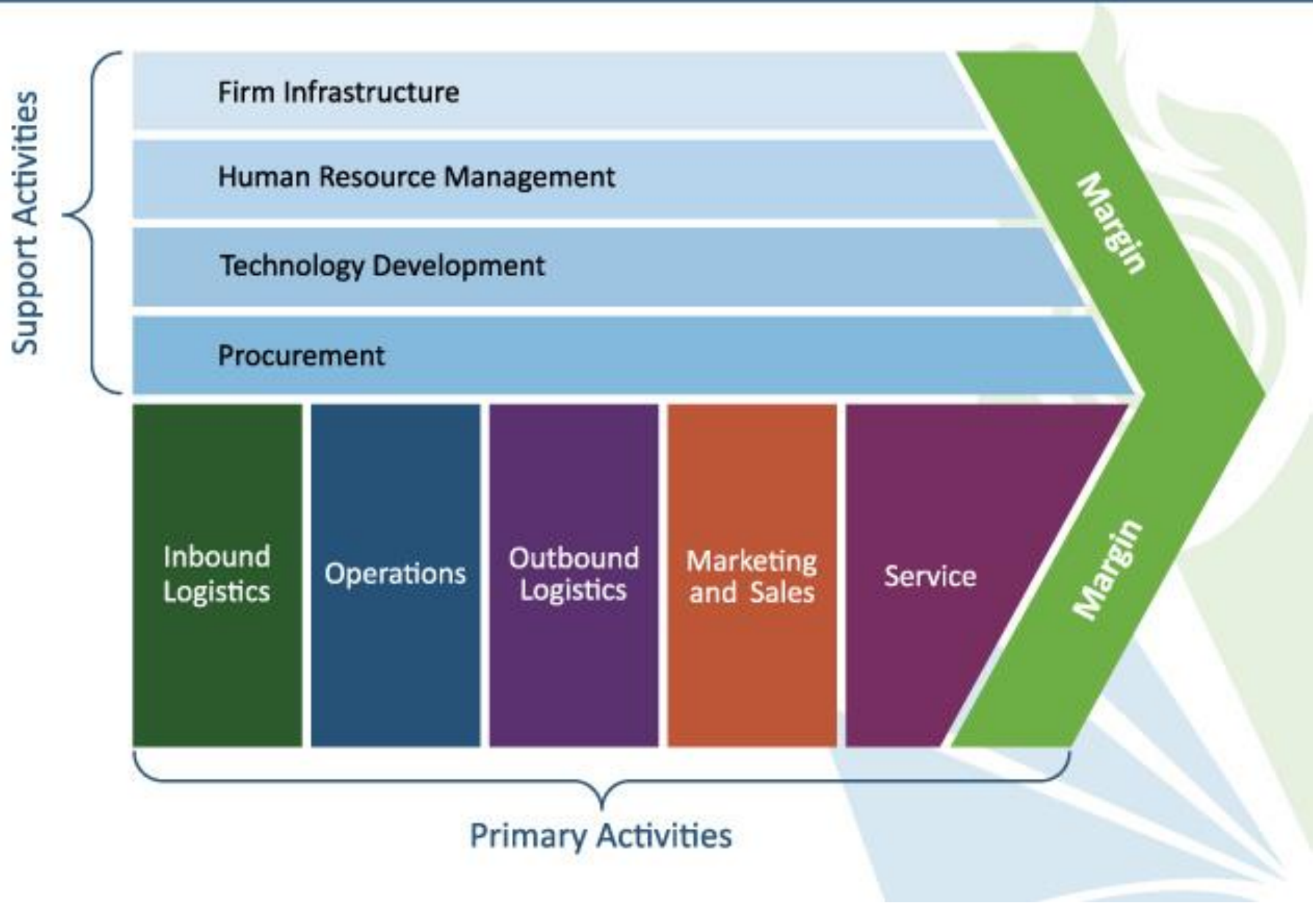
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- The goal of the organization is to undertake activities that create value for customers or other stakeholders.
- **Value chain:** Activities that increase the value of the goods or services sold to customers (i.e., these add customer utility to the goods and services).
- **Activities consume resources and cause costs.** Managers seek to identify activities that don't add value and eliminate those activities.
- Moving inventory from the factory to a warehouse is costly but may not add value.

Value Chain Analysis



- **Cost drivers:** underlying factors (variables) that cause costs. Understanding cost drivers allows us to control and manage costs.
- In Booth Executive Education, the number and variety of programs (custom v. open), the number of locations (Gleacher, London, Hong Kong, others), the number of instructors, and the number of participants enrolled all drive costs.
- The number of participants drives catering and printing costs; the number of locations drives faculty and staff travel costs as well as adding complexity to logistics related to program delivery.

- Our interest is in **costs**, not **expenses**.
- **Expenses** are defined in financial accounting (GAAP) as the cost of resources consumed in a period to generate revenue. GAAP rules determine when costs are expensed (e.g., rules for calculating/expensing depreciation, inventory costs, R&D costs).
- **Costs** are either **actual (cash) outlays** or **opportunity costs**. Outlays are usually cash outflows (e.g., the cost of raw materials purchases; wages paid to employees) that occur now or later. Accounting systems typically only capture actual outlays.

- The **opportunity cost** of a resource is the value of the resource in its most highly-valued *alternative* use (i.e., next-best use).
- Let's say we decide to use existing land, purchased 20 years ago, to build a new plant. What is its opportunity cost (to plug into the NPV calculation)?
 - GAAP measures the **historic cost** of the land: the cash outlay when the land was purchased. Not a good measure.
 - The land's current **net realizable value** (expected sales price net of transactions costs) is a better measure.
- Measuring opportunity cost depends on the **opportunity set**—the set of alternative uses of the resource.

- The amount spent on the land 20 years ago is a **sunk cost** that was incurred in past (cannot be changed). Sunk costs are unavoidable and non-recoverable.
- The cost to Airbus of developing the A380, at around \$25B, is sunk – unavoidable and non-recoverable.
- Sunk costs are **irrelevant for decisions** made today (because today's decisions are inherently forward looking).
- However, sunk costs may be relevant for other purposes – knowing how much was spent in the past may be useful for **control** (to hold managers accountable for past investment decisions, including M&A).

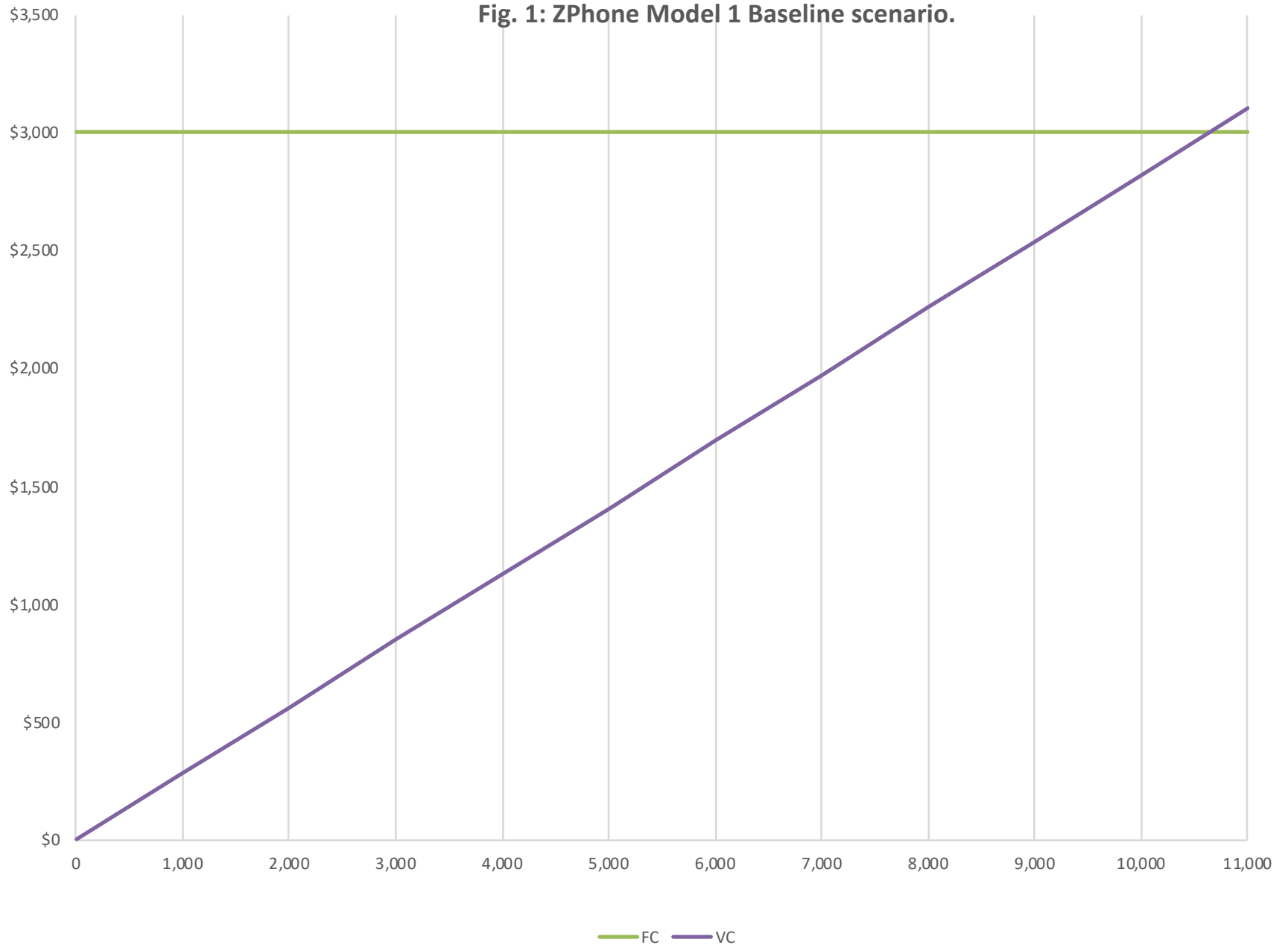
Fixed costs, variable costs, and the relevant range.

- An operations manager at American Airlines wants to know how costs will decline if the number of weekday flights from Chicago O'Hare to New York LaGuardia is cut from 12 to 6 – *an important question in March 2020*.
- To answer this, we need to know what costs are variable and what costs are fixed, *within the relevant range of activity*.
- A related but different question is the extent to which these costs are *avoidable* in a cash flow sense.

Variable costs: change in direct proportion to the level of activity.

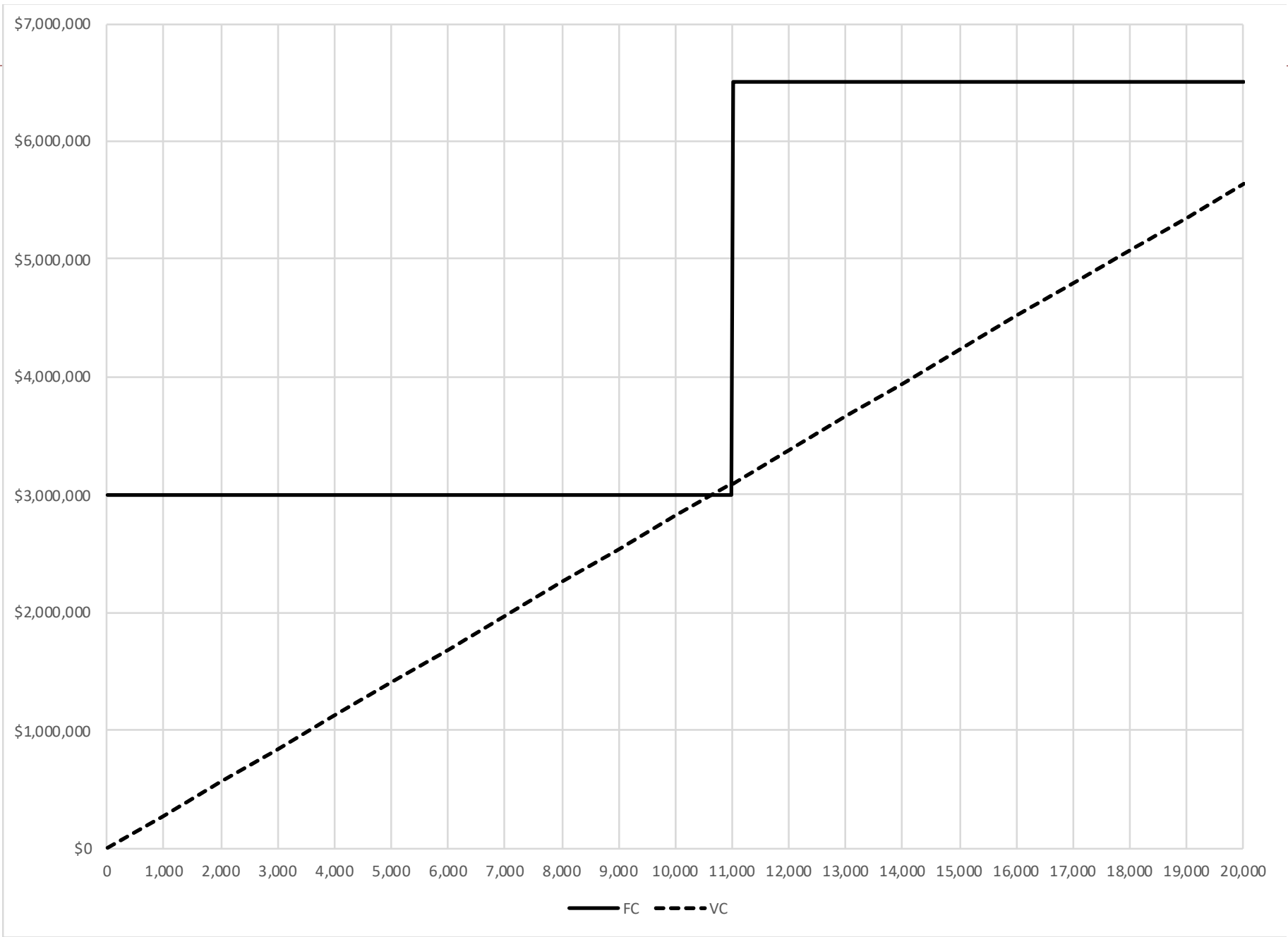
- Fuel costs are approximately proportional to the number of flight-miles, and so are variable. Crew costs depend on number of flights (their time is costed on an hourly basis) and so are also variable but with respect to flights (or flight-hours), a different activity measure.
 - Costs can only be variable with respect to a specific *cost driver* (*activity measure*).
 - Defining the right cost driver is critical. For airlines, is this number of passengers, number of flights, number of flight-miles, number of flight-hours, or something else? If we're interested in fuel costs, it depends on what best explains fuel consumption.

Fig. 1: ZPhone Model 1 Baseline scenario.



- **Fixed costs:** do not change in this range of activity. If we continue to operate flights to LGA from 6am-10pm each day, the cost of gate agents at LGA is fixed because we employ the same number of agents for the same hours regardless of the number of flights. The **cost of landing “slots” at LGA is also fixed**, as are equipment costs such as aircraft depreciation and lease rental payments – none of these change with the level of activity.
- **Over longer periods, most fixed costs become variable:** if we reduce the number of flights we can employ fewer gate agents each day and reduce labor costs. However, these costs may not be **avoidable** in a cash flow sense (we may still have to pay them).

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- Some costs are a combination of fixed and variable costs, including **step costs** and **mixed/semivariable costs**.
 - The cost of cleaning crews is a step cost if one crew can clean 7 aircraft in a four-hour shift (an additional crew is needed if there are more than 7 flights in this period).



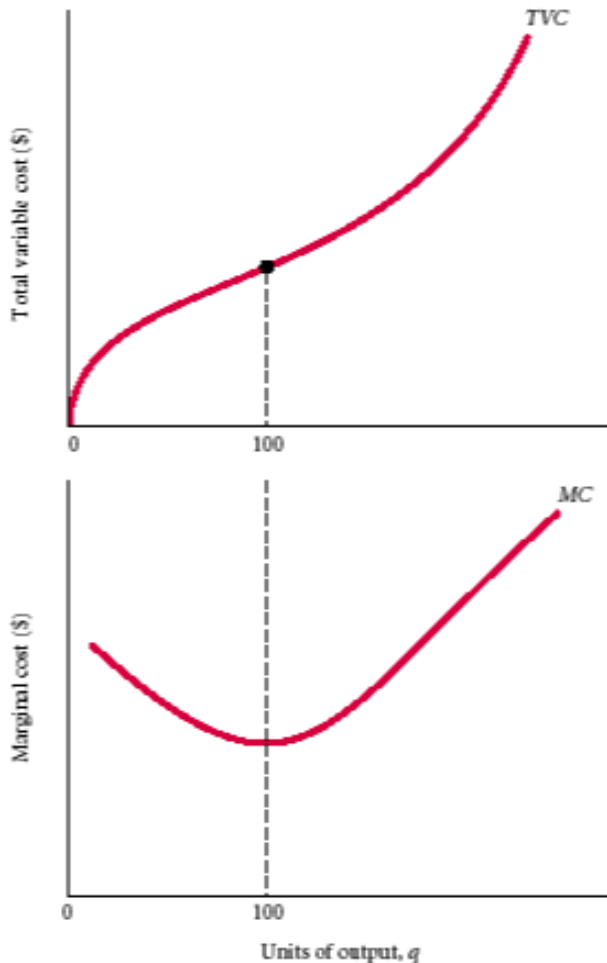
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- **Marginal costs:** the *incremental* cost of each additional unit of activity. These are unlikely to be constant over the relevant range of activity.

Common cost structure (i.e., production function):

Marginal costs are high at first, then decline, and then increase again.

Marginal cost: slope of the total cost curve at each point.

Variable costs: can approximate as the slope of a straight line fitted to the total cost curve where it is approximately linear (the relevant range).



- Just because costs are variable or fixed doesn't necessarily mean that they are **avoidable** (in the case of variable costs) or **unavoidable** (in the case of fixed costs).
- If we can't layoff employees who assemble products (direct, variable costs) those costs are not avoidable.
- Conversely, fixed costs may be avoidable. If we stop using a plant we may be able to avoid utilities costs (like electricity, heating) that are fixed over the range of activity relevant for that plant.
- Key idea: whether costs are **variable** or **fixed** does not say much about whether the costs are **avoidable**, which may be our key concern.